# CCRS File Processing Framework User Guide

## Version

This guide covers `Version 1.3.3`.

## Purpose

The CCRS File Processing Framework supports these workflows:

- process existing CCRS referral JSON files
- generate new referral JSON files from the bundled templates
- build a final `.gz` referral package
- process LEIE Excel files with strict duplicate detection
- upload referral or LEIE output to configured CCRS environments
- validate a completed LEIE load in the database and CCRS Table Viewer

## Install and Start the App

Download `CCRS.File.Processing.Framework.Setup.Version.1.3.3.exe` from the GitHub Releases page and run the installer.

The release contains a full Windows Setup installer. Portable executables are not generated.

The installer lets you choose the installation folder. Application files, bundled templates, assets, and documentation are installed with the app. Writable runtime data is stored under the current Windows user's AppData location so the app can run without administrator rights.

Runtime data includes:

- `config`
- `docs`
- `logs`
- `upload_import`

Use `GitHub Releases` for official installers. Build directories such as `dist-electron` and `release` are not part of the project or normal user workflow.

### What's New Release Notes

The app automatically displays the current release notes:

- the first time the app is opened after a new installation
- the first time the app is opened after upgrading from any older version

Select `Start Using The App` to acknowledge the notes. The same release notes will not open again on later launches, but a future release with a new release identifier will display its own notes once. This acknowledgement is stored in the current user's AppData so reinstalling or changing the installation folder does not confuse an older release with a newer one.

## Before You Begin

Prepare the items required for your workflow:

- referral processing: a folder containing CCRS `.json` files
- LEIE processing: an Excel file or folder containing `.xlsx`, `.xls`, `.xlsm`, or `.xlsb` files
- an output folder where processed files can be saved
- `WinSCP.ini` or a supported credentials JSON file if upload is required

## Main Areas

The app has three main tabs:

- `Referral Files`
- `LEIE Data Processing`
- `More Features Coming Soon`

`Referral Files` contains two modes:

- `Edit Existing Files`
- `Generate New Files`

## Workflow Progress and Large Files

The workflow stepper shows `Select`, `Configure`, `Validate`, `Process`, and `Upload`. The status card remains visible near the bottom of the application and includes a progress bar during

long operations.

File processing, LEIE work, gzip creation, upload tests, and uploads run in background workers so the window remains responsive. Select `Cancel` to stop a running process. A cancelled process can leave an incomplete file in the selected output folder; inspect or remove that file before restarting the same job.

Before processing, the source summary reports available file size and row information. The application protects the computer with configurable safety limits for:

- one source file and the combined source selection
- LEIE worksheet rows
- generated JSON file count
- referrals per generated file
- total referrals in one generated batch

If a source exceeds a safety limit, split it into smaller approved batches or ask the application maintainer to adjust the corresponding `CCRS_MAX_*` environment setting after reviewing available memory and disk space.

## LEIE Data Processing

Use `LEIE Data Processing` to inspect, clean, save, upload, and validate LEIE Excel data.

The `LEIE Data Upload and Validation Guide` button opens the separate detailed LEIE manual from the app.

### Step 1: Select the Source

Choose either:

- one LEIE Excel file
- a folder containing one or more LEIE Excel files

Supported extensions are `.xlsx`, `.xls`, `.xlsm`, and `.xlsb`.

The app scans the source and reports the number of files and rows it can read.

### Step 2: Resolve Duplicate Rows

LEIE duplicate detection always uses strict full-row matching. A row is considered a duplicate only when every column value matches another row after whitespace is trimmed and text is compared without case differences.

The duplicate area shows scan progress, the final result, and duplicate row details.

If duplicates are found, choose one action:

- `Remove duplicates automatically (keep first)` keeps the first matching row and removes later matches
- `Review and fix manually` blocks processing until the source is corrected and rescanned
- `Cancel Process` stops the LEIE workflow

Use `Rescan` after changing the source file manually.

## Step 3: Choose the Output

Select the output folder.

`Output File Name` defaults to the original source file name. The app does not add a `Processed_` prefix or otherwise change the source base name by default.

You may enter a different base name when needed. The saved output is an `.xlsx` workbook, and the live preview shows the final name.

## Step 4: Configure LEIE Upload

Enable `Upload the .xlsx file after processing` only when the processed workbook should be sent to one or more servers.

Supported targets are `DEV`, `PPD`, `SQA`, and `PROD`.

Every LEIE server defaults to this remote directory:

```
/in/manual_process/
```

This LEIE default is separate from the referral default `/CCRA_Uploads`. Switching workflows changes only a standard default path. A custom remote directory entered by the user is preserved.

For every selected server:

- choose the upload user
- confirm or edit the remote directory
- use `Test Connection` before processing when possible

## Step 5: Save and Upload

Select `Save Processed File` after the duplicate check is complete.

When upload is enabled, the app processes selected servers in order. For each server it:

- shows an `Uploading` state
- transfers the processed `.xlsx` file
- verifies that the remote file exists and has the expected byte size
- marks that server complete only after verification
- closes the connection with a bounded timeout so a stalled disconnect cannot block the workflow
- retries transient failures up to three times with increasing delay
- moves to the next selected server

If one server fails, the app records the error and continues with the remaining selected servers. The final status identifies failed targets. A successfully saved local workbook remains available even when an upload target fails.

## Step 6: Find the ETL Batch ID

After the server processes the uploaded file, use the latest batch log entry to identify the correct `etl_batch_id`.

The in-app example batch ID is:

```
LEIE07312026061530l3
```

Replace it with the actual batch ID for the environment and file being validated.

## Step 7: Run the LEIE Validation SQL

The LEIE Upload area contains an `ETL Batch ID` field and a `Validation SQL` box. Entering a batch ID updates all three batch-ID references automatically.

The SQL box uses color-coded SQL keywords, quoted values, numbers, and comments. It remains selectable, and `Copy SQL` copies the complete plain-text query set.

The current query set is:

```
Use ccrsdb;

select * from CLAIM_BATCH_LOG order by last_updated_date desc;--
First get the etl_batch_id

Select File_Name, *
```

```
from CLAIM_BATCH_LOG
where etl_batch_id = 'LEIE0731202606153013'
order by Start_date_time desc;

select count(*)
from dbo.LKUP_PROVIDER_SANCTION
where etl_batch_id = 'LEIE0731202606153013';

Select count(*)
from dbo.LKUP_PROVIDER_SANCTION
where etl_batch_id <> 'LEIE0731202606153013'
  and is_current = 'Y';

select * from dbo.LKUP_PROVIDER_SANCTION;
```

Run the SQL in `ccrsdb` for the selected environment. Confirm the file name, timestamps, batch status, loaded row count, and current-record state.

### Step 8: Validate in CCRS UI

Open the CCRS Table Viewer for the same environment, select `CLAIM_BATCH_LOG`, and confirm that the file name, `etl_batch_id`, timestamps, and load status match the database results.

Complete validation in one environment before moving to the next required environment.

## Edit Existing Referral Files

Use `Edit Existing Files` when referral JSON files already exist.

### Source

Choose the folder containing the input `.json` files. The app:

- reads JSON files and ignores unrelated files
- reports the number of source files found
- loads available defaults from the first source file
- displays the current source `ListDate`

### Referral Details

Set the starting referral number, `ListDate`, and any optional overrides. Preview lines update before the workflow runs.

### Output

Choose the output folder and optionally enter:

- the processed JSON base name
- the final `.gz` archive name
- archive additions such as `Network` and `Referral DateTime`

The app adds `.json` and `.gz` extensions automatically.

Examples:

- `batch101` becomes `batch101.json`
- multiple files become `batch101_1.json`, `batch101_2.json`, and so on
- `ccra_referrals_20260326` becomes `ccra_referrals_20260326.gz`
- archive additions can produce `customname_CCN1_20260401_134500.gz`

### Run

Select `Run Processing`. The app applies changes, creates processed JSON files, builds the final `.gz`, and uploads the archive when upload is enabled.

## Generate New Referral Files

Use `Generate New Files` to create referrals from the bundled templates.

Choose:

- the number of JSON files
- the number of referrals per file
- referral details and optional overrides
- output naming

This mode does not require a source folder and does not upload. Select `Generate Files` to run it.

# Referral Number and ICN

## Referral Number

Enter the user-controlled portion only. The app adds `VA` automatically and produces a 12-character final value.

Rules:

- letters and numbers are allowed
- the live preview shows the first value
- generated referral numbers remain unique within the run

Examples:

- `1` becomes `VA0000000001`
- `eldin` becomes `VA00000eldin`

## ICN

Open `Show Optional Fields` to configure ICN generation.

Available modes are:

- keep `TEST` and generate the remaining characters
- keep `TEST` and enter a custom alphanumeric suffix
- enter a custom full ICN base and generate the remaining characters

Generated ICNs remain unique. Choose `3`, `4`, or `5` to make the ending of each ICN match the ending of its Referral Number.

# Patient Identity Controls

## SSN and DOB

Random generation is enabled by default. Each patient receives a different random test SSN and random DOB. Every generated SSN starts with `999`; only the remaining six digits change.

Uncheck the corresponding random option only when a fixed value must be applied. Fixed SSNs must also begin with `999` and use the displayed `999-99-0001` format. Dates must use `YYYY-MM-DD`.

## Station ID

Choose an exact `station_key` from the reference-backed dropdown or enable random valid Station IDs.

The dropdown uses the actual `DIM_VA_STATION.station_id`. For example, Tremont VA Clinic uses `614GA01`; the former catalog row number `2242` is not a Station ID and is no longer accepted or generated.

Rules:

- a selected or entered Station ID is never padded or extended
- a three-character Station ID remains three characters
- valid VA Station IDs may contain three through seven letters and numbers
- manual values must exactly match the reference table
- automatic mode shuffles only the valid entries from `config/station_options.json`
- automatic IDs change between referrals without repeating until the reference list is exhausted
- the shipped catalog is refreshed during app upgrades

## SEOC Number and SEOC ID

Select either a SEOC Number or a SEOC ID. The other selector updates automatically so the referral receives a valid synchronized pair from the bundled reference data.

The following special pairs are marked `Special: explicit selection only`. The bundled generator never assigns them automatically:

| SEOC key | SEOC ID |
|---|---|
| `67` | `INF_FERTILITY-1332 EXISTING OVARY_1.1.17` |
| `158` | `EUC_EMERGENCY CARE_1.2.8` |
| `191` | `EUC_POM USE ONLY UNAUTHORIZED EMERGENCY ROOM CARE_1.1.7` |
| `233` | `SSC_TRANSPLANT BONE MARROW COMP EVAL_1.5.14_REV` |
| `268` | `MHC_MENTAL HEALTH RESIDENTIAL TREATMENT_1.15.7` |
| `1160` | `MHC_RESIDENTIAL WITHDRAWAL MANAGEMENT_1.22.2` |

To use one, select its key or ID. The matching value is filled automatically. Leave both selectors at `Keep source SEOC` when no explicit override is required.

All dropdowns use the same compact design as Station ID. Referral / ICN Match, Network, SEOC, and upload-user lists remain inside a scrollable menu instead of expanding beyond the application window. Long selected labels are shortened visually without changing their stored values.

## Other Optional Referral Fields

Supported optional fields include:

- Payment Authority Code
- Network
- Referral Category
- Referral Date
- Referral Date Time
- Referral Expiration Date
- Referral Last Update Date
- Referral Period
- Referral Consult ID

Payment Authority Code accepts up to five numeric digits. Later generated referrals increment the value so it remains unique.

Network values are `CCN1`, `CCN2`, `CCN3`, `CCN4`, and `CCN5`.

If an optional field is blank, the app keeps the source value unless that field's selected mode explicitly generates a value.

## Referral Upload

Referral processing uploads the final `.gz` archive. Enable `Upload the .gz file after processing`, select one or more targets, choose the user, and confirm the remote directory.

The standard referral remote directory is:

```
/CCRA_Uploads
```

This location is specific to CCRS Referral uploads. LEIE Data uploads use the separate default `/in/manual_process/`.

Target toggles remain disabled until the main upload option is enabled.

## Upload Credentials

The app uses runtime copies of:

- `config\upload_profiles.json`
- `config\upload_credentials.json`

Use `Import To Config` to import `WinSCP.ini` or a supported credentials JSON file. The app does not automatically search arbitrary machine locations for credentials.

Importing WinSCP settings reads the saved sessions, resolves servers and users, and writes the active credentials configuration under the app's runtime data location.

On Windows, imported passwords are encrypted using operating-system-protected Electron safe storage. Existing plaintext credential files are migrated automatically when the application can access Windows encryption. The original JSON or INI file selected for import is preserved.

Each upload target includes `Test Connection`. Use it to verify the selected account and remote directory before a full workflow.

## Validation and Status Messages

The app validates fields before running and displays messages next to invalid inputs.

Validation includes:

- SSN `999` prefix and format
- Referral Number format
- Station ID reference match
- Payment Authority Code range
- date and date-time format
- generator counts
- selected upload user
- remote upload directory
- LEIE duplicate resolution

Common statuses include:

- `File Loaded`
- `Duplicates Found`

- `Ready to Save`
- `Uploading`
- `Uploaded`
- `Upload Complete`
- `Upload Warning`
- `Failed`

During a long operation, the progress bar reports the current file or server. `Cancel` remains available even though editing controls are temporarily disabled. Referral and LEIE uploads report each server independently and continue through later selected servers after an individual failure.

## Diagnostic Report

Select `Export Diagnostics` in the application header and choose a folder. The generated JSON report contains the application version, runtime versions, processing limits, non-sensitive workflow preferences, reference-data counts, and upload server/user counts. It does not contain passwords, credential values, input paths, output paths, or source file contents.

## Referral Output Format

The final `.gz` contains one combined JSON object:

```
{
  "ListDate": "YYYY-MM-DD hh:mm:ss",
  "Referrals": [
    {}
  ]
}
```

## Files and Cleanup

The packaged app opens `docs\USER_GUIDE.pdf` when the User Guide is requested.

General behavior:

- generated JSON files use lowercase `.json`
- one `.gz` archive is created per referral-processing run
- existing `.gz` files are not deleted automatically
- processed JSON files may be kept or removed after archive creation

- app-created staging folders and temporary upload helper files are cleaned after a run or when the app closes
- release binaries are distributed through GitHub Releases rather than stored in the source repository

# Troubleshooting

## LEIE Output Name Is Unexpected

Select or rescan the source. The default should match the original source base name. Clear a custom value if you want the original name restored.

## LEIE Upload Appears Stuck

Check the individual server status. The app verifies the remote file and limits disconnect waiting time. If a server reports an error, review its user, path, and connection test; later selected servers are still attempted.

## A Large File Is Rejected

Review the displayed file size or row count. Split the source into smaller approved batches. Do not raise a safety limit without confirming that the workstation has enough memory and disk space.

## A Job Was Cancelled

Check the output folder for an incomplete workbook, JSON file, or gzip archive. Remove incomplete output before running the same job again.

## Upload Target Is Disabled

Enable the main upload checkbox first, then enable each required server.

## No Upload Profiles Are Available

Use `Import To Config` and select the approved `WinSCP.ini` or credentials JSON file.

## LEIE Processing Is Blocked

Resolve duplicates automatically, or correct them in the source and select `Rescan` before saving.

## SQL Uses the Example Batch ID

Paste the actual `etl_batch_id` into the in-app field before selecting `Copy SQL`.